Ethan’s thesis to your CFD-to-1D thesis

Evidence-backed integration plan | 20 October 2026

Review edition. 24 integration proposals, 22 fact checks, 61 numbered equations, 31 supplemental equation groups, 33 captioned figures and 39 embedded image resources.

The highest-value integration is to improve the provenance and interpretation of your computational reference: reconcile configuration versions, distinguish the PIV measurement operator from conservative transport, strengthen wall and energy verification, and qualify uncertainty claims. Modern HITEC properties, annular heat loss and recirculation motivation are largely already present in your manuscript.

Scope: This report is a reviewable plan. Neither your manuscript nor production model has been modified. Proposed changes require an agreed scope. Numerical checks diagnose the supplied PDF; they are not reruns of Ethan’s unavailable code.

Page convention: Source locations use PDF page indices, starting at 1, plus printed thesis page numbers. For the body and appendices, printed page = PDF page - 14. The title page says May 2026; PDF creation metadata is later and does not determine the degree date.

Controlling comparison: CSEM_Master_s_Dissertation (7)(2).zip, current six-chapter source. Chapters 2-5, air-property appendix, bibliography, README and AGENTS instructions were inspected. No executable simulation archive was supplied. The review therefore gives code behavior and tests instead of inventing exact runtime module paths.

Recommended order

1. Resolve configuration, property and observation-operator provenance before transferring numbers.

2. Make documentation and citation corrections; keep frozen results and model definitions intact.

3. Implement independently testable wall, enthalpy and uncertainty diagnostics after locating the exact runtime source.

4. Test one physical variant at a time, with fixed electrical inputs and a declared calibration/test split.

5. Keep water turbulence and pipeline replacement outside the present core scope.

Configuration crosswalk

Pipe ID

Your source: 0.870 in=22.098 mm

Ethan: 0.930 in=23.622 mm, body / A.1; A.2 later says radius11.049 mm

Implication: 14.27%area difference; loss and Re change; direct calibration transfer invalid.

Quartz ID / length

Your source: 1D21.209 mm; CFD~20.94 mm; length185.93 mm

Ethan: mesh bore21.082 mm from2*10.541; length186.0 mm

Implication: Close but distinct geometry records; retain both.

Slope/path

Your source: idealized36 in legs, connectors; total148 in in prior loop definition

Ethan: 21.5 deg reconstructed incline;4.353m path in Table 5.1

Implication: Need actual coordinate/arc/junction lengths; do not infer commonloop by case names.

Cooling length

Your source: 304.8 mm activejacket

Ethan: nominal1ft;A.1coolersegment262.5 mm plusreducers

Implication: Active/nominal heated-area ambiguity must be resolved.

Insulation

Your source: CFD1.40 in Calzite; baseline1D reference1 in times empirical factor; localized full1.40 in

Ethan: salt2.00 in,water0.40 in effective; ceramic manufacturer fit

Implication: Different setup and fitted definitions; do not change ouralpha_R by copying thickness.

HITEC

Your source: cp1423.47; exponentialmu; tabulated CFD transport/analyticROM

Ethan: main modern suite matches; B.1legacycp/mu conflicts

Implication: Keep modern package and reconcile actual simulation coefficients.

Mesh/model

Your source: ~2.17M cells,laminar

Ethan: coarse2.552M,medium8.241M,fine28.614M; selected water SSTLM variants

Implication: Source verification uncertainties do not transfer to our mesh.

Sensor QOI

Your source: signed net transport and declaredTP/TW reductions

Ethan: PIVproxy/best wall azimuth/meanTP,wallTW10jacket

Implication: Match observation operators before comparing errors.

Detailed integration proposals

P0 denotes a claim or provenance blocker; P1 denotes high-value integration; P2 denotes optional work or work outside present scope. These are proposals, not silently accepted edits.

I01 | P0 | Configuration/provenance gate

Source: 4.1.1 PDF 47-49 / print33-35;5.1.1 PDF 63-65;5.7 Table 5.1 PDF 81;A.1-A.3 PDF 121-124

Attribution: Ethan reconstruction; supplier parts, photographs and reported private correspondence, not survey-grade experimental truth

Insight: Nominal0.93 in pipe ID,21.5 deg incline,0.83 in test bore,4.353m path,2.00 in salt insulation. Cooler nominal1ft differs from262.5 mm active mesh segment. Appendix calls right leg riser although body calls it downcomer.

Compatibility: Our Ch3 table loop_summary uses0.87 in bore,20 deg-style idealization,0.835 in 1D quartz bore vs20.94 mm CFD,304.8 mm jacket,1.40 in Calzite. At fixed mass flow density,0.93 in has14.27% larger area and12.49% lower velocity; fixed-flow laminar pipe resistance drops23.41%. These are scaling comparisons, not predicted loop changes.

Proposed change: Ch3: add a versioned geometry/material crosswalk. Code: import separate configuration profiles, never overwrite frozen CFD/ROM setup. Record original input hashes and mapping of all junctions/heat patches.

Validation: Recover actual mesh/dictionaries behind both studies, compute area/volume/centerline lengths, compare to records and rerun only an isolated geometry sensitivity. Gate any quantitative transfer on reconciliation.

Status: Required before numerical adoption

I02 | P0 | Property package and dimensionless labels

Source: 4.3.1 Eqs4.2-5 PDF 52-54;5.2.2 Eqs5.15-18 PDF 72;B.1 TableB.1 PDF 125;C.2 TableC.3 PDF 131

Attribution: Legacy property laws cited to Janz/Reis; modern viscosity/cp from Jin/Parida; reconstruction and choice are Ethan's

Insight: Body uses modern exponential mu and1423.47cp, appendix gives legacy polynomials. Published Pr labels31.74-54.29 do not describe modern property package unchanged. Appendix global/local Pr rows also disagree.

Compatibility: Our Ch4 eq:hitec-* already matches body modern package; changing to TableB.1 would regress consistency. Our tabulated CFD transport vs analytic ROM also needs interpolation parity.

Proposed change: Ch2/3/4: identify historical case labels versus recomputed local/global nondimensional numbers and temperature definitions. Code: one versioned property registry plus explicit legacy_reproduction profile; no retuning of frozen P1/P2.

Validation: Compare property values and derivatives at measured and solver temperatures; sweep positivity and range; test Kelvin conversion and tabulated interpolation; record units and composition basis. W1/W2/W12.

Status: Required audit; modern package already present

I03 | P0 | Cylindrical wall heat transfer

Source: 3.4 PDF 40-43 Fig 3.1/Eq 3.12;5.4.2 Eqs5.23-25 PDF 77-78

Attribution: Classical annular conduction and radiation; custom rc External Temperature implementation is Ethan's

Insight: Per-inner-area resistance sum r0 ln(ri/ri-1)/ki and external area ratio rout/r0 are essential for thick insulation; no axial wall conduction/storage.

Compatibility: Our Ch2 cylindrical resistances and Ch4 localized wall states already capture key idea. Ethan's solid material and thickness values differ, so transfer equations/verification strategy rather than coefficients.

Proposed change: Add explicit area basis and source partition to Ch4; audit implementation q_in*Ain=q_out*Aout. Clarify wall-face vs cell-center Tp. Extend candidate BC tests, not replace current closures.

Validation: Constant-k multilayer cylinder, convection-only/radiation-only/mixed, zero source, heater-layer source and limiting cases; compare integrated powers and dimensional residuals. Source p77.

Status: Recommended; partial existing implementation

I04 | P0 | Air cooler enthalpy correction

Source: 5.1.3 Eq 5.6 PDF 68 / print54;B.3 TableB.4 PDF 127

Attribution: Ethan formula; enthalpy identity is standard thermodynamics

Insight: cp(Tout)Tout-cp(Tin)Tin is not integral cp dT. AppendixB coefficients over25-to113C give+2.1408% error; midpoint-cp approximation differs-0.0292%.

Compatibility: Our Ch4 cooler uses effectiveness-NTU and air balance, not automatically the same bug. Inspect actual ROM code before changing it.

Proposed change: Document and implement exact polynomial enthalpy integral where variable cp is used; retain current effectiveness-NTU model and geometry. Add a separate comparator reproducing printed formula for diagnosis only.

Validation: Constant-cp limit; analytical polynomial primitive vs quadrature; invariance of enthalpy difference to reference offset; air/fluid cooler heat balance. W3 supports identity.

Status: Recommended conditional correction

I05 | P0 | Experimental energy consistency

Source: 4.5.1 Eqs4.6-8 Tables4.4 PDF 55-56;C.1 TableC.1 PDF 130

Attribution: Reis equations/measurements as transcribed; negative-loss reconstruction is Ethan's

Insight: Table 4.4 reports negative inferred salt losses for1-3 under legacy cp; table DeltaT row conflicts with TP6-TP2 from C.1. Corrected-cp diagnostic with reported mdot and C.1 gives208.27,238.43,226.19,194.56 W. Those are recalculations, not new measured heat inputs.

Compatibility: Our electrical input37 W test section and gross/net wall flux ledger are already explicit. Do not impose this inferred net heat or silently discard cases from our CFD assessment.

Proposed change: Add Ch3 experimental-data caveat and separate measurement table with original/reconstructed columns. Code: control-volume-specific energy ledger; keep electrical vs net deposited power and measurement operators separate.

Validation: Reconstruct all eight cases from raw temperatures, powers, mdot processing and cp; propagate full uncertainty; compare reported values without overwriting originals.

Status: Required before citing energy paradox as experiment failure

I06 | P0 | PIV observation operator versus transport

Source: 3.5 PDF 43-44;4.5.2 Eqs4.9-10 Table 4.5 PDF 56-57;6.2 PDF 97;6.10 PDF 109

Attribution: Experimental measurement belongs to Reis; right-half axisymmetric reconstruction and<1.3% reproduction are Ethan's

Insight: Axisymmetric integral from a single planar half-profile is an observational reconstruction; three-dimensional asymmetry/recirculation violates symmetry. Table 6.10 explicitly uses piv_method.

Compatibility: Our Ch2 signed mass-flux and net-enthalpy reductions are more appropriate for conservative ROM runtime. Keep them; experimental-proxy agreement does not establish same net loop mdot.

Proposed change: Ch3: label mdot_net, mdot_forward, mdot_reverse and mdot_piv_proxy separately, with density/area/plane/time averaging and symmetry assumptions. Code: optional observation-operator module outside runtime targets.

Validation: Apply both operators to identical full CFD fields; integrate signed flux on multiple planes, assess right/left proxy bias and recover raw published PIV where available.

Status: Recommended; do not replace conservative targets

I07 | P0 | Quantity-specific verification

Source: 3.1-3.3 Eqs3.1-11 PDF 35-40;6.2 Tables6.6-9 PDF 93-98;A.4 PDF 124

Attribution: Richardson/GCI from Roache/Celik; two-grid application is Ethan's

Insight: Full-loop study completed only coarse/medium; cannot estimate observed p or establish monotonic/asymptotic convergence from two solutions. Table 3.1 cell targets differ from A.3 by2x in bulk; mesh counts2.552/8.241/28.614M differ from our2.17M reference.

Compatibility: No imported GCI values apply to our production mesh. Our Ch3 already states mesh-evidence limitations; preserve them.

Proposed change: Add source-specific limitations to Ch3 and Ch5. Code/report: GCI per exact QoI/mesh family with realized h, assumed vs observed p, iterative error and stationarity flags; unavailable estimates remain missing, not zero.

Validation: Three settled meshes of same geometry/BC and reduction; grid-independent probes/planes; temporal/iterative error small relative to differences. If only two, publish assumed-order estimate with Fs and assumptions, not demonstrated order. W5/W6.

Status: Required claim boundary

I08 | P0 | Uncertainty and confidence

Source: 3.6 Eq 3.15 PDF 44;5.7 Tables5.1/2 PDF 81-85;6.3 PDF 97-98;7.1 PDF 113

Attribution: ASME framework is cited; input bounds/propagation/99% claim are Ethan's choices

Insight: Uncertainties combine standard input distributions with GCI safety bands; independence and covariance require care. No shown coverage factor establishes99%. Insulation-k and air-flow uncertainty omitted; high fit R2 does not bound extrapolation error.

Compatibility: Our retrospective ROM-to-CFD assessment must not become99%-validated experiment claim by inheritance. Shared CFD/experiment inputs can correlate uncertainties.

Proposed change: Ch5: quantitative uncertainty budget per QoI, standard vs expanded vs bounds and provenance; first-order Jacobian covariance or sensitivity ensembles in separate analysis scripts. Replace generic validated language with error and interval definitions.

Validation: Check independent synthetic cases, correlated sensors/shared properties, nonlinear ensemble sensitivity, and withheld cases. W4/W5/W14.

Status: Recommended; no99% claim

I09 | P1 | Strong sensor-position sensitivity

Source: 3.5 PDF 43-44;4.2.1 PDF 50;6.4 Figs6.10-12 PDF 99-103

Attribution: Experimental nominal probes from Reis; CFD spatial-sensitivity result is Ethan's

Insight: 0.1 in (2.54 mm) neighborhood around TP3 can change reported temperature by>10 K. Wall azimuth is uncertain; TW10 is jacket exterior.

Compatibility: Directly relevant to our probe projections and existing TP2 sensitivity; cannot calibrate internal h to a misregistered sensor.

Proposed change: Ch3: sensor observation-map table including fluid centerline, steel outer wall, jacket wall; augment location and azimuth sensitivity near gradients. Code: fixed coordinate/region operators and uncertainty ensembles, never select best location using test data.

Validation: Sample resolved fields in physically justified coordinate bounds, compare local/bulk temperatures, freeze mapping before scoring; report nominal and marginalized errors.

Status: Recommended

I10 | P1 | Recirculation and thermal stratification

Source: 6.4 Figs6.10,6.16-20 PDF 102,105-108

Attribution: Ethan CFD observation; later experimental confirmation is reported private correspondence, not independently documented raw observation

Insight: Fast near-wall rising stream and reverse flow in opposite region persist into quartz/test section; complex tee geometry deflects the stream.

Compatibility: Supports our existing two-state momentum/recirculation motivation. It does not identify exchange coefficients or prove a thermal-exchange model improves held-out targets.

Proposed change: Ch3/5: cite source for qualitative mechanism, retain our measured forward/reverse metrics and conservative two-state model. New thermal closure only as hypothesis with same-QoI identification.

Validation: Recompute signed cross-section mass and enthalpy budgets, momentum leverage and reverse area on our fields; qualify at own meshes/windows; test withheld powers without refit.

Status: Recommended explanatory integration

I11 | P1 | Developing-flow length scales

Source: 2.1 PDF 26-27;3.4.1 PDF 41;6.1 Eqs6.1-2 PDF 87;3.13 PDF 43

Attribution: Straight-tube textbook estimates; Ethan applies them to this facility

Insight: Hydrodynamic and thermal entrance lengths have different coefficients for different development tolerances. In verification Re334/Pr53.2/D20 mm gives0.334m and17.77m with0.05 coefficient.

Compatibility: Our Shah entry friction and Muzychka-Yovanovich heat transfer are already developing-flow choices. Junctions, recirculation and heating resets mean no universal single entry coordinate.

Proposed change: Ch2/4: state entry-length definition, tolerance, reset convention and applicable flow assumptions; compare Graetz/Sellars reference only on a straight-tube benchmark.

Validation: Fully developed asymptotes; entrance-limit behavior; laminar reference data; never substitute3.657 everywhere or apply Dittus-Boelter at these Re. W7.

Status: Recommended clarification

I12 | P1 | External convection and orientation

Source: 5.1.2 Eqs5.1-5 PDF 67-68

Attribution: Churchill-Chu correlations are cited; sin(theta)^2 blend is Ethan's interpolation

Insight: Vertical plate Nu_L and horizontal cylinder Nu_D use different characteristic lengths. Eq 5.3 writesD generally; combining Nu values would be ambiguous, whereas5.5 blends h.

Compatibility: Our ambient model can use same family, but must check outer diameter, segment length, film air properties, curvature and ambient conditions.

Proposed change: Ch2/4: cite both1975 papers, define characteristic length per correlation, expose blend as assumption. Code: test h_horizontal=Nu_D k/Dout and h_vertical=Nu_L k/L before blending.

Validation: Orientation endpoints0/90 deg, unit checks, film-property sweep, vertical-cylinder criterion and sensitivity to blend exponent/ambient; do not call interpolation universally validated. W8/W9.

Status: Recommended audit

I13 | P1 | Fins/stubs and missing heat paths

Source: 5.1.4 PDF 69 / print55;5.7 PDF 82-84

Attribution: Composite fin model and9-16% inferred loss share are Ethan's; geometry partly assumed

Insight: Seven cap/valve/off-gas paths modeled as2D axisymmetric solid-core composite fins; solid-core model can overestimate hollow/valve losses.

Compatibility: Relevant to our junction/stub ambient loss terms and localized heat-path model, already present. Percentage and fin conductances cannot transfer across different geometry/insulation.

Proposed change: Add a physically grounded fin candidate to separate structural-loss class, leaving current calibrated coefficients/frozen results intact. Record cap/valve lengths and hollow sections.

Validation: Compare analytic fin limits,2D energy conservation, hollow vs solid sensitivity, and independent surface temperatures; identify loss using dedicated data rather than aggregate temperature fit.

Status: Conditional candidate

I14 | P1 | Heater and quartz source distribution

Source: 4.1.1 PDF 48;6.5 PDF 110-111

Attribution: Ethan interpretation of experiment; localized heating evidence cited to Reis2024 heater-failure article[24]Fig 8

Insight: Uniform quartz heating may miss two end rings; local near-wall heat can alter viscosity and velocity-profile development.

Compatibility: Our setup uses37 W on quartz. Same total electrical input does not imply same spatial source or deposited-fluid heat.

Proposed change: Ch4/5: label current uniform distribution; propose ring/end-patch sensitivity with preserved total power. No coefficient adjustment until input/source geometry reconciled.

Validation: Use documented ring locations/widths or explicit bounds, integrate source to37 W, compare profiles/temperature/heat ledger and held-out cases; do not infer rings solely from improved fit.

Status: Conditional experiment

I15 | P1 | BC verification benchmark independence

Source: 3.4 Eq 3.12/13 PDF 40-43;6.1 Tables6.2-5 PDF 86-92;C.3 PDF 132

Attribution: Ethan benchmark uses analytical resistance and empirical developing-film correlation

Insight: Mode A/B/C separate convection/radiation. Fine reference errors~0.05-0.07%; however equation3.13 at Gz3554 gives Nu23.7336 while appendix says26.67. Geometry polygonization raises integrated-Q observed order~2.74-2.98.

Compatibility: Useful test architecture, not evidence our BC implementation/mesh is verified. Our actual layer heat sources and temperature-dependent material treatment need additional tests.

Proposed change: Use independent pure conduction boundary tests with prescribed inner wall temperature first, then fluid-coupled benchmark; repair reference Nu, area and mesh-size bookkeeping before using tolerances.

Validation: Separate analytic geometry, meshed geometry and per-area flux vs integrated power; test residual, solution difference and refinement. Do not mistake algebraic qbalance machine precision for model accuracy.

Status: Recommended benchmark redesign

I16 | P1 | Nonlinear solver and steady-state evidence

Source: 5.1.2 PDF 68;5.5.1 PDF 78-79;5.6 PDF 80-81

Attribution: Ethan implementation choices

Insight: Under-relaxation0.5; wall iterations delta Ts<1e-3 K/deltaq<1e-2 W/m2. CFD pseudo-time Co_max5, terminal100s statistics; no physical transient accuracy claimed.

Compatibility: Our production source states different stationarity windows/tolerances and solver schemes. We can adopt diagnostic philosophy, not those numerical thresholds blindly.

Proposed change: Keep own stationarity gate. Add normalized global energy and loop momentum residuals, root-bracket diagnostics and iteration sensitivity to reduced solver; document actual pseudo-time scheme.

Validation: Tighter tolerances, different initial guesses/brackets and multiple terminal windows; timestep sensitivity if fluctuations affect means. Do not interpret solver fluctuations as physical transition without temporal checks.

Status: Recommended diagnostics

I17 | P2 | Water transition modeling

Source: 5.2.3 Eq 5.19 PDF 73-74;6.2 PDF 94;6.4 PDF 102,106;Table 6.10 PDF 109

Attribution: Ethan exploratory SSTLM/Gb variants; original Reis article reports laminar boundary-layer scaling

Insight: Text predictswater3/4 benefit, actual selected best fits usewater2Gb andwater4SSTLM. Global Ra>1e10 alone is not a universal pipe-loop transition criterion. Local y+ reaches~5.

Compatibility: Current work is laminar HITEC-focused; water turbulence is outside main comparison and cannot justify changing salt closure.

Proposed change: Keep in literature/limitations only. If later extending to water, declare model selection before validation and implement buoyancy source with correct rho normalization.

Validation: Independent transition data, model coefficient/source audit, near-wall resolution and actual-time/timestep sensitivity; compare all cases without selecting model post hoc. W10/W12.

Status: Defer outside present scope

I18 | P1 | Digitized-profile provenance

Source: 4.5.3 Figs4.2-4/Tables4.6/7 PDF 58-61

Attribution: Ethan extraction and salt raw-data cross-check; water raw full fields unavailable

Insight: Masking/calibration/brightness classification, column median and smoothing; global1.36% RMSE rounded1.5%, but one height3.65% RMS and14.10%max.

Compatibility: Useful if raw PIV cannot be obtained; no need if our target is own CFD fields. Noise/crossing smoothing can bias profiles.

Proposed change: Add digitization workflow only for external experimental comparison; store source image, calibration points, smoothing and profiles with pointwise quality flags.

Validation: Overlay digitized result, blind validation on available raw salt data, per-case/per-height uncertainty and endpoint sensitivity. Do not apply1.5% as guaranteed pointwise bound.

Status: Conditional supporting analysis

I19 | P0 | Salt 1 evidence handling

Source: 6.5 PDF 110-111;TableC.1 PDF 130;Table 6.10 PDF 109

Attribution: Ethan suspects label/entry/configuration mismatch; not a demonstrated correction to original experiment

Insight: Salt 1 mean-fluid error-24.43 K and proxy-mdot error-28.18%. TW10 Salt 1 exceeds Salt2 by0.36 K, smaller than stated1.56 K difference uncertainty; that observation alone cannot establish faulty data.

Compatibility: Our Salt 1 CFD development case is distinct from a questionable experiment label. Do not remove or reclassify our case by name alone.

Proposed change: Ch3/5: mark suspected experimental inconsistency unresolved; report with/without sensitivity only when same underlying measurements are identified. Preserve all original values and labels.

Validation: Trace run IDs/powers/heating/airflow/insulation to raw experimental records; compare uncertainty-normalized differences; no confirmed fault statement without documentary evidence.

Status: Required wording restriction

I20 | P0 | Result interpretation and calibration

Source: Table 6.10 PDF 109;6.4 PDF 99,106;5.7 PDF 82;7.1 PDF 113

Attribution: Ethan results and post hoc choices

Insight: Seven non-Salt 1 aggregate case verdicts lie inside adopted bands. Wall values use best-aligned azimuth; water2/4 use selected turbulence models; water insulation effective 0.40 in was adjusted for fit. These are retrospective comparisons.

Compatibility: Our thesis title already says retrospective model assessment, and target-informed P2-CFD is identified. Ethan's table does not upgrade our CFD-driven ROM to experimental predictive validation.

Proposed change: Ch3/5/6: separate calibration/reconstruction, solution verification, experimental comparison and ROM-to-CFD assessment. Cite Table 6.10 as reported under its assumptions, not validated universal truth.

Validation: Freeze configuration/measurement operator/model before withheld data; recompute per-probe/profile errors, coverage and case aggregates; no best-of-model scoring presented as prospective performance. W5.

Status: Required claim boundary

I21 | P1 | Insulation and material properties

Source: 4.1.2 PDF 49;5.7 PDF 82-84;B.4 TablesB.5 PDF 127,unnumbered k PDF 128

Attribution: Ethan manufacturer-fit and extrapolation; SS/quartz engineering constants

Insight: Ceramic fit differs from our Calzite/ceramic input; appendix typo6.4209E-0.2; aluminum epsilon0.10 in Table 5.1 vs0.15 inB.5. Insulation-fit R2=0.998 does not establish room-temperature accuracy.

Compatibility: Do not replace our temperature-dependent SS316L/current insulation coefficients with Ethan's constant16.3 W/mK steel or different ceramic fit.

Proposed change: Ch4: cite actual manufacturer/material identity and fit range, record extrapolation uncertainty. Code: registry with provenance/validity, exact coefficient digits, source-specific emissivity.

Validation: Sweep actual wall/insulation temperatures; compare source datasheet points, positivity, uncertainty and sensitivity. Require new evidence before changing material package.

Status: Recommended provenance audit

I22 | P2 | Automated CFD pipeline and mesh quality

Source: 5.3/5.6 PDF 74-81;A.3 TablesA.4/5 PDF 124

Attribution: Ethan foam-vv/classy_blocks/Jinja orchestration; software utilities are cited

Insight: Structured conformal O-grids, coarse-to-medium mapping and diagnostic extraction aid reproducibility. Hexahedral mesh is not by itself proof of smaller error than every polyhedral alternative.

Compatibility: Reusable workflow pattern; actual source repo and YAMLs are not included in PDF or our manuscript ZIP.

Proposed change: Request/locate exact foam-vv commit/config files when implementation begins. Add manifest/provenance/quality-gate design; do not rebuild framework from prose without necessity.

Validation: Reproduce one case from frozen manifest; mesh quality checks plus QoI sensitivity and parallel consistency.

Status: Defer code adoption until source available

I23 | P0 | Bibliography and source version

Source: Title PDF 1;references[48][50][51] PDF 119-120;our references.bib keyrozak2026cfd Natural Circulation

Attribution: Direct bibliographic comparison

Insight: Uploaded title begins High-Fidelity CFD Simulation... and title page says May2026; our bib entry uses reversed title,aug and Forthcoming. PDF metadata creation Aug2026 does not prove degree/title month. Santini entry prints1948 instead of1984; Jin author list omits Xuehui An.

Compatibility: Safe documentation edits once selected; do not invent repository DOI/URL or claim published status.

Proposed change: Update existing Rozak key to exact supplied title, year, and note that supplied manuscript title page says May2026; confirm month before bibliographic finalization. Add distinct Churchill-Chu horizontal entry. Verify all reused original-source metadata.

Validation: Bib TeX uniqueness/build, titles/DOIs against primary publishers, citation resolution and scientific-claim guardrails.

Status: Recommended documentation change

I24 | P1 | Preserve model hierarchy and agreement gates

Source: Ethan7.2 PDF 113;comparison to our Ch4/5

Attribution: Integration judgment based on both sources

Insight: Much of Ethan's radial physics, modern salt properties, energy ledger and recirculation motivation is already in our manuscript. Highest value is provenance, operator and uncertainty repair.

Compatibility: Retain P1, P2-baseline naming preference, P2-CFD target-informed scale and frozen train/test split. Source ZIP still usesP2-unity in one table; user's later preference controls name when editing is agreed.

Proposed change: Implement selected documentation changes first, observation/verification tooling second, physical variants last. No numeric tables updated until reproducible reruns are available.

Validation: Run repository compile_and_qa and affected scientific guardrails; reproduce baseline results before/after code changes and report any altered result explicitly.

Status: Plan only; awaiting agreed scope

Numerical evidence and limits

Table 6.10, PDF 109 / printed 95, reports the following aggregate comparisons. All mass flows here use piv_method, not a three-dimensional signed mass-flux operator. Wall comparisons use a best-aligned azimuth. Water 2 selects SSTLM_Gb; Water 4 selects SSTLM; remaining cases use laminar results. Reported rounded values are retained in the machine-readable CSV.

Salt 1 | laminar

Mass flow: prediction 0.011347 kg/s, reported reference 0.0158 kg/s; error -28.18%, adopted validation uncertainty 5.20%. Mean fluid-temperature error -24.43 K with uncertainty 3.80 K. Wall error -17.07 K with uncertainty 3.80 K. Ethan’s aggregate verdict: No.

Salt 2 | laminar

Mass flow: prediction 0.016311 kg/s, reported reference 0.0168 kg/s; error -2.91%, adopted validation uncertainty 7.50%. Mean fluid-temperature error -2.27 K with uncertainty 3.96 K. Wall error +0.67 K with uncertainty 3.84 K. Ethan’s aggregate verdict: Yes.

Salt 3 | laminar

Mass flow: prediction 0.018466 kg/s, reported reference 0.0175 kg/s; error +5.52%, adopted validation uncertainty 9.77%. Mean fluid-temperature error -1.04 K with uncertainty 4.28 K. Wall error +1.48 K with uncertainty 4.06 K. Ethan’s aggregate verdict: Yes.

Salt 4 | laminar

Mass flow: prediction 0.021974 kg/s, reported reference 0.0201 kg/s; error +9.32%, adopted validation uncertainty 9.63%. Mean fluid-temperature error -2.20 K with uncertainty 4.75 K. Wall error +0.88 K with uncertainty 4.34 K. Ethan’s aggregate verdict: Yes.

Water 1 | laminar

Mass flow: prediction 0.0085476 kg/s, reported reference 0.0083 kg/s; error +2.98%, adopted validation uncertainty 8.53%. Mean fluid-temperature error +1.36 K with uncertainty 1.77 K. Wall error +1.03 K with uncertainty 1.77 K. Ethan’s aggregate verdict: Yes.

Water 2 | SSTLM_Gb

Mass flow: prediction 0.010188 kg/s, reported reference 0.01 kg/s; error +1.88%, adopted validation uncertainty 15.17%. Mean fluid-temperature error +0.60 K with uncertainty 1.93 K. Wall error +0.56 K with uncertainty 1.93 K. Ethan’s aggregate verdict: Yes.

Water 3 | laminar

Mass flow: prediction 0.012166 kg/s, reported reference 0.012 kg/s; error +1.38%, adopted validation uncertainty 7.15%. Mean fluid-temperature error +0.74 K with uncertainty 2.05 K. Wall error +0.93 K with uncertainty 2.05 K. Ethan’s aggregate verdict: Yes.

Water 4 | SSTLM

Mass flow: prediction 0.015055 kg/s, reported reference 0.0151 kg/s; error -0.30%, adopted validation uncertainty 7.85%. Mean fluid-temperature error +1.37 K with uncertainty 2.38 K. Wall error +1.59 K with uncertainty 2.38 K. Ethan’s aggregate verdict: Yes.

Seven case labels are reported within adopted bands, but the result is conditional retrospective agreement, not a demonstration of general or prospective accuracy. Aggregate mean temperatures do not establish pointwise temperature agreement. These numerical errors do not transfer to your differently configured CFD reference.

Other reported evidence: composite fins account for 9-16% of modeled heat removal (Section 5.1.4, PDF 69); a 0.1-inch-radius location neighborhood can change salt probe temperatures by more than 10 K (Section 6.4, PDF 99/103); finest-grid integrated power differs from the boundary reference by 0.062-0.067% (Tables 6.2-6.4, PDF 90-91), subject to the reference-Nusselt inconsistency below.

Fact-check register

Verification is bounded. Primary publisher and official NIST, ASME and OpenFOAM sources were used. Several full texts were paywalled or blocked; these checks are explicitly partial. This review does not certify every literature statement or simulated result in the thesis. A printed mistake does not establish that the unavailable code used it, and a correct printed formula does not establish that the code implemented it.

F01 | Confirmed PDF error

Location: 4.3.1 PDF 54 print40

Claim: 200C is493.15 K

Finding: 200+273.15=473.15 K. Listedrho1938.88,mu0.00677,k0.547 match473.15 K, not493.15 K. At493.15 K:rho1923.885445,mu0.00549763,k0.552677576.

Action: Correct nominal-temperature label; do not change valid underlying formulas.

Evidence: E; direct arithmetic

F02 | Confirmed mathematical error in printed formula

Location: 5.1.3 Eq 5.6 PDF 68

Claim: cp(Tout)Tout-cp(Tin)Tin is an improved variable-cp enthalpy difference

Finding: dh/dT=cp, whereas d(cp*T)/dT=cp+T*dcp/dT. At298.15-to386.15 K using TableB.4, integral=88816.3876 J/kg; printed=90717.7736 J/kg; midpoint=88790.4720 J/kg.

Action: Use integral and check actual source code before concluding simulations used printed mistake.

Evidence: E; W3; direct integration

F03 | Confirmed internal conflict

Location: 5.2.2 PDF 72 vsB.1 PDF 125

Claim: AppendixB.1 lists coefficients used in simulations

Finding: Maincp1423.47/exponentialmu conflict with legacy quadratic cp/cubic mu inB.1. Exact simulated package cannot be established solely from contradictory PDF.

Action: Use executable physical Properties dictionaries/commit; do not copy appendix blindly.

Evidence: E; A; W1; W2

F04 | Confirmed internal arithmetic mismatch

Location: Eq 3.13 PDF 43 vsC.3 PDF 132

Claim: Gz3554 gives Nu26.67

Finding: Stated formula evaluated at Gz3554 yields23.7335613453; h=Nu*0.540/0.020=640.8062 W/(m2 K), rather than719.8. Reference Rint becomes0.00156053m2 K/W rather than0.00139.

Action: Recompute reference benchmark from actual Gz definition/temperature/length; do not infer full BC failure from this alone.

Evidence: E; W7; direct arithmetic

F05 | Method correction required

Location: 4.5.1 Eq 4.6 PDF 55

Claim: Uncertainty for difference Qnet=Qh-Qloss is Qnet timesRSS(relative uncertainties)

Finding: For independent inputs variance=sigma Qh^2+sigma Qloss^2. With correlation subtract2cov(Qh,Qloss). Qloss inferred from Qnet is correlated; use primitive measurements to avoid circular propagation.

Action: Use Jacobian covariance and full measurement model; distinguish source transcription from own corrected method.

Evidence: E; W4

F06 | Confirmed unsupported confidence interpretation

Location: 7.1 PDF 113;Eq 3.15 PDF 44

Claim: Validated within99% confidence interval

Finding: No corresponding coverage factor/coverage derivation was identified; RSSstandard uncertainties are not inherently99%. GCI safety factor is not a confidence quantile.

Action: Report stated uncertainty components/coverage separately; do not repeat99%.

Evidence: E; W4; W5; W14

F07 | Confirmed two-grid limitation

Location: 6.2 PDF 93;Tables6.6-9 PDF 97-98

Claim: Full-loop convergence follows3-grid method

Finding: Text explicitly says only two settled refinements used. Assumed order required; no observed order or convergence classification can be estimated from two values.

Action: Label assumptions and do not import GCI numbers to unrelated2.17M mesh.

Evidence: E; W6

F08 | Confirmed table mismatch

Location: Table 3.1 PDF 40 vsA.3 PDF 124

Claim: Full-loop bulk cell targets

Finding: Table 3.1=2.25/1.50/1.00 mm;A.3=1.125/0.750/0.500 mm. First-cell values agree. Mesh resource/radii entries also differ across appendices.

Action: Resolve against actual block Mesh Dict/check Mesh before reproducing.

Evidence: E

F09 | Primary-source corroboration, bounded

Location: Eq 5.17 PDF 72

Claim: Base HITECcp=1423.47 J/(kgK)

Finding: RSC primary indexed text reports1.42347 J/(g C), equivalent1423.47 J/(kgK). This is a sample measurement, not universal exact material truth or quantified4%standard uncertainty.

Action: Keep source/version/composition and realistic uncertainty; no need to change our matching package.

Evidence: E; W1

F10 | Partial external verification

Location: Eq 5.16 PDF 72

Claim: Jin viscosity law/range/2.5% uncertainty

Finding: Primary publisher confirms study authors/year and2.5% reported uncertainty. Full fitted coefficients and valid-range statement could not be read in original article during this run.

Action: Retain coefficients as source-transcribed; obtain primary article/full data before declaring fully independently verified.

Evidence: E; W2

F11 | Confirmed citation error

Location: Reference48 PDF 119

Claim: Santini conductivity paper published1948

Finding: Publisher identifies1984,volume27(4),623-626,DOI10.1016/0017-9310(84)90034-6.

Action: Use correct metadata; exact correlation/range still source-limited.

Evidence: E; W11

F12 | Modeling assumption, not law

Location: 5.1.2 Eq 5.5 PDF 68

Claim: sin(theta)^2 blend predicts inclined-cylinder convection

Finding: Correct endpoints do not validate interior angles or vertical-cylinder curvature; blend ofh must retain each length scale.

Action: Sensitivity study and provenance; add horizontal-cylinder citation absent from citedvertical entry.

Evidence: E; W8; W9

F13 | Confirmed case-label inconsistency

Location: Table 4.1/4.2 PDF 53/55 vsC.3 PDF 131

Claim: One set of salt Pr values

Finding: 4.2 gives54.29,50.81,41.65,31.74;C.3global gives54.29,54.07,53.89,53.63. Modern package at200C gives Pr17.61496. Different properties/location definitions cannot be mixed.

Action: Recompute from declared T/mu/cp/k; preserve historical labels as labels.

Evidence: E; A; direct arithmetic

F14 | Confirmed temperature-difference mismatch

Location: Table 4.4 PDF 56 vsC.1 PDF 130

Claim: DeltaT isTP6-TP2

Finding: Table 4.4salt DeltaT=11.16,10.09,7.95,5.63 K;C.1differences=9.26,9.97,9.08,6.80 K. Table values may use different definitions, but not declared consistently here.

Action: Trace measurement definitions; do not claim negative losses robust to property/operator change.

Evidence: E; direct subtraction

F15 | Unsupported categorical diagnosis

Location: 6.5 PDF 110

Claim: Salt 1TW10>Salt 2 proves faulty experiment

Finding: Difference116.88-116.52=0.36 K, below1.56 K stated difference uncertainty. Other mismatches warrant investigation, not proof of data corruption.

Action: Use suspected unresolved inconsistency; preserve raw cases.

Evidence: E; W4; arithmetic

F16 | Confirmed table/verdict qualification

Location: 6.10 PDF 109;6.4 PDF 106

Claim: Seven validated cases demonstrate general predictive accuracy

Finding: Errors reported are aggregate means/proxyflows; best wall azimuth and selected turbulence models were used. Retrospective effective water insulation0.40 in is fit-informed.

Action: Use bounded comparison wording and report per-probe/profile diagnostics, setup dependence and calibration.

Evidence: E; W5

F17 | Confirmed overgeneralization risk

Location: 5.2.3 PDF 73

Claim: RaL>1e10 proves turbulence/justifies SSTLM

Finding: No universal loop transition criterion is established. Original Reis abstract reports laminar wall scaling; simulation best-fit model selection is not turbulence observation.

Action: Do not infer salt turbulence or replace our laminar setup.

Evidence: E; W10

F18 | Confirmed weaker-than-claimed bound

Location: 4.5.3 Table 4.6 PDF 61

Claim: 1.5% digitization uncertainty bounds all water profile points

Finding: Assigned fromglobal1.36%RMS;worst salt-heightRMS3.65%,Linf14.10%. Transferring to water is an assumption.

Action: Case/height and pointwise uncertainty; preserve smoothing/crossing bias.

Evidence: E

F19 | Confirmed notation/units issue

Location: Eq 5.13 PDF 71

Claim: alpha_eff is thermal diffusivity

Finding: mu/Pr=k/cp haskg/(m s), not m2/s. Kinematic thermal diffusivity is k/(rho cp).

Action: Name/units must follow enthalpy equation convention.

Evidence: E; dimensional analysis; W12

F20 | Confirmed bibliographic mismatch

Location: PDF 1 and ourreferences.bib

Claim: Existing Rozak bib matches supplied thesis

Finding: Existing title differs,monthaug,note Forthcoming; supplied title page says May2026 and exact title High-Fidelity CFD Simulation of Single-Phase Natural Circulation Loop Experiments: Verification and Validation.

Action: Update after agreed scope, retain existing citation key; final degree month/status needs confirmation.

Evidence: E; A

F21 | Confirmed internal typo/parameter conflicts

Location: B.4 PDF 128;Table 5.1 PDF 81 vsB.5 PDF 127

Claim: Insulation polynomial and Al emissivity unambiguous

Finding: Second duplicate fit prints6.4209E-0.2 instead of coherent6.4209e-2; prose says polynomial does become negative despite preceding intended positive behavior. Al epsilon0.10 vs0.15.

Action: Use first coherent formula only for diagnostic; validate material datasheet and actual source inputs.

Evidence: E

F22 | Confirmed acceptance-summary inconsistency

Location: 6.1.4 PDF 92 vs Table 6.5 PDF 91

Claim: AllGCIfine<0.01%

Finding: Integrated-QfineGCIs forM/F/X are0.0101-0.0104%, slightly greaterthan0.01%;fluxGCIs are0.0013-0.0018%. Both are far below stated2% threshold.

Action: Correct summary wording and specify QoI; distinction does not by itself overturn benchmark.

Evidence: E

Most important equations for our work

The complete Equation_Inventory.html and CSV/JSON files include all 61 numbered equations, source-page contexts, readable La TeX, units, attribution and priority. Supplemental unnumbered groups include nondimensional definitions and every property-coefficient family. Source-page images preserve unclassified inline algebra. Priority A means essential to audit or integrate, not automatically approved for production.

3.12; 5.23-5.25: Annular resistance, inner/outer area normalization and nonlinear wall balance. Essential to audit thick insulation and electrical source partition.

5.7; 4.7-4.8: Global and segment energy accounting. Essential to distinguish prescribed electrical input, net fluid heat, passive losses and structural paths.

4.9-4.10: PIV observation reconstruction. Essential because a planar axisymmetric proxy is not the same operator as conservative net transport.

3.1-3.11; 3.14-3.17: Quantity-specific mesh verification, error and uncertainty. Essential, with explicit assumptions for two-grid estimates and coverage.

5.15-5.18: Modern HITEC properties, already matched in your source. The conflicting legacy appendix must not override them.

3.13; 6.1-6.3: Straight-tube developing-flow benchmark and entrance scales. Useful verification reference under its own assumptions; the appendix Nu must be reconciled.

5.1-5.5: External convection and radiation. Keep correct characteristic lengths and identify the inclination blend as an engineering assumption.

5.6: Do not adopt as printed. Use mdot times the integral of cp over temperature for an enthalpy difference.

Figures and image extraction

Image_Atlas.html links all 33 captioned figures, every one of the 39 page-associated native raster resources, and 139 rendered pages. Each native resource has a page, xref, dimensions and hash. Alpha masks are composited where present. Vector plots are preserved in the original PDF and individual source-page PDF exports. Figure-context PDFs include captions and any neighboring figure; they are not mislabeled as isolated raster originals.

Highest value: Figures 6.10-6.11 (sensor sensitivity), 6.16-6.20 (three-dimensional recirculation), 3.1 (annular circuit), 5.1-5.2 (geometry provenance), and 6.3 (PIV plane). Prefer your own corresponding plots for manuscript figures and cite Ethan for the mechanism or method. Facility photographs/experimental PIV retain Reis attribution. Appendix C captions are repetitive and should not be used alone to identify plot content.

Implementation sequence

Documentation first: proposed scope is Chapters 03/04/05 and references.bib, with a configuration crosswalk and exact source locations. Preserve CFD comparison versus experiment and retrospective versus predictive claim boundaries. The later user preference P2-baseline controls the former P2-unity label; do not alter P2-CFD’s target-informed status.

Reproduce before modifying: recover exact runtime commits, dictionaries, geometry and property tables, probe/plane operators, time windows and baseline reports. Add diagnostics before changing physics.

Physical candidates: separate quartz end-ring heating, structural fin losses, sensor positions and convection assumptions. Identify one change at a time against relevant quantities, then hold parameters fixed on withheld cases.

Numerical acceptance: conservation residuals, correct limiting behavior, dimensional consistency, matched operators and quantity-specific verification. Use expanded uncertainty only with a declared coverage model.

Manuscript acceptance: follow repository scope/instructions, build and run the scientific QA scripts after approved changes. No tables or claims are updated from imagined reruns.

Unresolved evidence

The cause of Salt 1 discrepancy; original PIV processing scripts; wall-probe azimuth/location; quartz ring heating; as-built insulation and material identity; actual code and coefficients behind conflicting appendices; assumed full-loop two-grid convergence order; and exact source correlation uncertainty/ranges where full texts were inaccessible. Codex should retain these as unresolved rather than inventing inputs.

Sources and verification scope

E | Ethan Rozak thesis, supplied PDF

Checked: Primary supplied source; all 139 pages extracted. Equations and important disputed values checked in page renders.

Scope: Thesis reports, not independent reproduction of simulations.

A | Andres thesis source: CSEM_Master_s_Dissertation (7)(2).zip

Checked: Current supplied Library snapshot read locally; chapters 2-5, appendices, bibliography and repository instructions inspected.

Scope: Controls current model definition; source is not an executable CFD/ROM run archive. No manuscript edits made.

W1 | Parida & Basu (2023), RSC Advances 13, 5496-5508

Checked: Primary publisher indexed text verified cp=1.42347 J/(g K), weak liquid-phase temperature dependence. Full HTML/PDF retrieval returned403.

Scope: Supports base-salt cp measurement; does not establish Ethan's +/-4% as standard uncertainty.

Open primary source

https://pubs.rsc.org/en/content/articlehtml/2023/ra/d2ra07384f

W2 | Jin et al. (2016), Accurate viscosity measurement of nitrates/nitrites salts for concentrated solar power

Checked: Primary publisher indexed abstract/metadata verified authors, study and 2.5% reported uncertainty; full article inaccessible.

Scope: Exact exponential coefficients and420-695 K bounds not independently verified from full article in this run.

Open primary source

https://www.sciencedirect.com/science/article/abs/pii/S0038092X16303759

W3 | NIST Chemistry Web Book, Shomate cp and enthalpy expressions

Checked: Primary NIST indexed equations checked; cp polynomial and enthalpy integral have different coefficient factors.

Scope: Thermodynamic identity check; not a source of the actual air mixture coefficients.

Open primary source

https://webbook.nist.gov/cgi/cbook.cgi?ID=C7732185&Mask=15

W4 | NIST TN1297, Appendix A: Law of Propagation of Uncertainty

Checked: Full official page read, especially A-3 covariance terms and A-7 rectangular distribution.

Scope: Supports derivative-based propagation, covariance and a/sqrt(3).

Open primary source

https://www.nist.gov/pml/nist-technical-note-1297/nist-tn-1297-appendix-law-propagation-uncertainty

W5 | ASME V&V20-2009 (R2021), official scope

Checked: Full official product scope read. Complete standard not accessible.

Scope: Validation is variable- and validation-point-specific; extrapolation beyond those points is beyond its stated scope.

Open primary source

https://www.asme.org/codes-standards/find-codes-standards/standard-for-verification-and-validation-in-computational-fluid-dynamics-and-heat-transfer/2009

W6 | Celik et al. (2008), Procedure for Estimation and Reporting of Uncertainty Due to Discretization in CFD Applications

Checked: Primary publisher metadata found; full text403. Method details are source-transcribed and algebraically checked, not full-standard-certified.

Scope: Citation and methodological provenance for GCI; exact procedure needs authoritative full text for implementation.

Open primary source

https://asmedigitalcollection.asme.org/fluidsengineering/article/130/7/078001/444689/Procedure-for-Estimation-and-Reporting-of

W7 | Lienhard & Lienhard, A Heat Transfer Textbook, 6th ed. (2024), Eq 7.29

Checked: Author textbook PDF search-index equation matches Ethan3.13; direct web open failed.

Scope: Corroborates formula form; no claim of independently verifying its quoted <1% error over every condition.

Open primary source

https://bpb-us-e1.wpmucdn.com/sites.mit.edu/dist/6/1810/files/2024/04/AHTTv600.pdf

W8 | Churchill & Chu (1975), horizontal cylinder free convection

Checked: Primary publisher indexed title/abstract and DOI verified; complete equations not readable from publisher in this run.

Scope: Correct separate provenance for horizontal-cylinder correlation, not vertical-plate paper alone.

Open primary source

https://www.sciencedirect.com/science/article/pii/0017931075902227

W9 | Churchill & Chu (1975), vertical plate free convection

Checked: Citation metadata corroborated; formula in source checked algebraically. Full original not accessed.

Scope: Applicability to slender vertical cylinders and an inclination blend remains a separate model assumption.

Open primary source

https://doi.org/10.1016/0017-9310(75)90243-4

W10 | Reis, Seo & Hassan (2024), Physics of Fluids36,033605

Checked: Primary publisher abstract and references read; experimental body/data not available in retrieved page.

Scope: Original report attributes near-wall salt velocity peaks to developing thermal layers and reports laminar boundary-layer scaling. Detailed table values here are Ethan's transcription, not independently checked raw experiment.

Open primary source

https://pubs.aip.org/aip/pof/article/36/3/033605/3268730/Molten-salt-flow-visualization-to-characterize

W11 | Santini et al., Measurement of thermal conductivity of molten salts in the range100-500 C

Checked: Primary publisher indexed metadata establishes1984; Ethan bibliography prints1948.

Scope: Bibliographic correction only. Exact correlation uncertainty/range not verified from full original.

Open primary source

https://www.sciencedirect.com/science/article/abs/pii/0017931084900346

W12 | OpenFOAM Foundation/CFD Direct v13 thermophysical models

Checked: Official guide read: he Rho Thermo, energy selection and property layers.

Scope: Supports configuration vocabulary; exact custom BC and coded turbulence source require actual source files.

Open primary source

https://doc.cfd.direct/openfoam/user-guide-v13/thermophysical

W13 | Eca & Hoekstra (2014), numerical-uncertainty procedure

Checked: Author/institution paper located and abstract retrieved.

Scope: Potential alternative for non-asymptotic/noisy mesh convergence; not automatically executed.

Open primary source

https://www.marin.nl/api/marin/downloads/get-download/5675/0/860/block_files_utb8kwxda60a/VsLN4atol Tpe BkST8w5DtHMRBwg%3D/2014-JCP-Vol262-Eca_Hoekstra-V%26VProc.pdf

W14 | NIST TN1297, reporting uncertainty

Checked: Official indexed guidance checked for probability interpretation and coverage reporting.

Scope: A claimed confidence level needs a stated coverage model/factor; RSS alone does not supply99%.

Open primary source

https://www.nist.gov/pml/nist-technical-note-1297/nist-tn-1297-7-reporting-uncertainty